

The associations between traumatic experiences and trajectories of substance use in adolescence and young adulthood – the role of acute neuroendocrine and subjective stress reactivity

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ABSTRACT

Adolescents exposed to traumatic events, including physical and sexual abuse, are at higher risk of developing problematic substance use behaviours. This study investigates associations between traumatic experiences and trajectories of substance use in adolescence and young adulthood, focusing in particular on differences in neuroendocrine and subjective stress reactivity as potential explanatory mechanisms. Using data from the TRacking Adolescents' Individual Lives Survey (TRAILS) (N = 715), we assessed whether traumatic experiences up to age 16 were associated with differences in acute neuroendocrine stress reactivity (in terms of heart rate (HR), high-frequency heart rate variability (HF-HRV), pre-ejection period (PEP), and cortisol reactivity), as well as subjective stress reactivity, assessed in the context of a standardized stress test conducted around age 16. We then analysed whether these stress reactivity measures predicted substance use trajectories (tobacco, alcohol, and cannabis) from around age 16 to around age 22. Results showed that traumatic experiences increased the risk of following trajectories characterized by an early initiation and escalation of tobacco, alcohol, and cannabis use (vs. no use or low use trajectories). Traumatic experiences were additionally associated with a higher probability of belonging to cannabis use trajectories characterized by consistently relatively low levels of use (vs. no use) or a later escalation of use in young adulthood. Regarding mechanisms, blunted neuroendocrine stress reactivity, but not subjective stress reactivity, was inconsistently linked to trajectories of higher substance use. However, differences in acute stress reactivity did not contribute to associations between traumatic experiences and substance use trajectories, as associations between traumatic experiences and acute stress reactivity were absent. Our findings emphasize the need to explore mechanisms beyond acute stress reactivity that may explain the association between trauma and adolescent and young adult substance use.

1. Introduction

Exposure to traumatic events, such as domestic violence, sexual abuse, and natural disasters, is strongly associated with increased risks of substance use and substance use disorders (SUDs) in adolescents and young adults (Kirsch and Lippard, 2022). Traumatic experiences may affect substance use through various pathways, including neurobiological mechanisms (al'Absi et al., 2021) and changes in how individuals react to stressful events, potentially leading to the onset or escalation of

substance use (al'Absi et al., 2021; Chaplin et al., 2018). Although there is no universally accepted definition that distinctly separates traumatic events from other forms of adversity, it is generally acknowledged that traumatic events are characterized by the highest levels of severity (Krupnik, 2019).

Many studies have examined, and sometimes shown, associations between traumatic experiences and both hyper- and hypo-reactivity of the hypothalamic–pituitary–adrenal (HPA) axis and the two branches of the autonomic nervous system (ANS) (i.e., the sympathetic nervous

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system (SNS) and the parasympathetic nervous system (PNS)), key neuroendocrine systems that help the body cope with stress, in adolescence (Kuhlman et al., 2015; Trickett et al., 2014). Fewer studies have investigated associations between traumatic experiences and subjective stress (SS) reactivity (i.e., the degree to which an acute stressor triggers negative emotional states), revealing mixed results (Hengesch et al., 2018; Trickett et al., 2014). Other studies have found associations between both hyper- and hypo-reactivity of the HPA axis and ANS and increased risks of harmful substance use behaviours in adolescents and young adults (Chaplin et al., 2018), whereas associations between SS reactivity and substance use have emerged to be inconsistent (Evans et al., 2012, 2013; Rahal et al., 2023b). However, the extent to which atypical neuroendocrine or subjective stress responses act as explanatory mechanism within associations between traumatic events and problematic substance use in adolescence and young adulthood is unclear. This study aimed to directly address this question.

1.1. Associations between traumatic experiences and atypical stress responses in adolescents

Studies assessing associations between traumatic experiences and neuroendocrine stress responses in adolescents have found divergent results, with differences depending on age, type and timing of traumatic events, and the neuroendocrine measure under consideration (an overview of key neuroendocrine stress systems and measures of acute neuroendocrine and subjective stress reactivity is provided in Table S1). For example, interpersonal violence exposure was associated with HPA-axis and SNS hypo-reactivity, specifically blunted cortisol and pre-ejection period (PEP) responses to a laboratory stressor, but not PNS reactivity (i.e., high-frequency heart rate variability (HF-HRV) reactivity) (Busso et al., 2017). Associations between maltreatment and blunted cortisol reactivity have also been found, particularly for survivors of sexual and/or physical abuse (Trickett et al., 2014) and in early but not late adolescence (Peckins et al., 2015). These findings are not ubiquitous, with some evidence that physical abuse is associated with cortisol hyper-reactivity (Kuhlman et al., 2015) or null associations (Zimmerman et al., 2020). Blunted HR-reactivity (reflecting both SNS and PNS responses) has been associated with childhood maltreatment in adolescents and young adults (Ji et al., 2024; Zhang and Lü, 2022). Finally, despite evidence that young adults who were adopted after being institutionalized showed greater SS reactivity compared to controls (Hengesch et al., 2018), several studies found no differences in SS reactivity between maltreated and non-maltreated adolescents (Marques-Feixa et al., 2023; Trickett et al., 2014).

Several meta-analyses that focused on early life adversity (ELA) in general, rather than traumatic experiences specifically, report mostly support for the hypothesis that ELA is associated with blunted cortisol, HR, and PEP reactivity (Brindle et al., 2022; Bunea et al., 2017; Schär et al., 2022; Young-Southward et al., 2020). Results regarding HF-HRV reactivity were less conclusive, with one meta-analysis finding an association between ELA and blunted HF-HRV reactivity only amongst individuals who were older and/or experienced adversity less recently (Wesarg et al., 2022). Lastly, a meta-analysis that distinguished between severe and less severe ELA found that less severe forms of adversity (e.g., parental conflict and discrimination) were associated with *heightened* cortisol reactivity, whilst traumatic events (e.g., physical and sexual abuse) were associated with cortisol *hypo-reactivity* (Hosseini-Kamkar et al., 2021). Under severe chronic stress or trauma perceived as unavoidable, it may be adaptive for the body's stress physiology to become hyporeactive, prioritizing energy conservation. In contrast, moderate adversity, which is more manageable, may prompt neuroendocrine adjustments that enhance energy mobilization during acute stress, consistent with hyperreactivity (Hosseini-Kamkar et al., 2021). These findings highlight the importance of distinguishing between less severe adversities and traumatic events (e.g., severe early adversity).

1.2. Associations between atypical neuroendocrine and SS reactivity and adolescent and young adult substance use

Neuroendocrine and psychological hyper- and hypo-reactivity to stress may increase risks of substance use in adolescents and young adults (Chaplin et al., 2018). Adolescents with hyper-reactivity may use substances to cope with hyper-arousal and increased negative emotions, whereas adolescents with hypo-reactivity may suffer from chronic under-arousal and may use substances to up-regulate emotional experiences and positive sensations (Chaplin et al., 2018). Several studies have found associations between blunted neuroendocrine stress responses and adolescent substance use, for example in terms of cortisol (Marceau et al., 2020; Marceau et al., 2022; Rahal et al., 2023b), HF-HRV (Rahal et al., 2023a), PEP (Rahal et al., 2023a), and HR reactivity (Evans et al., 2012, 2013). However, associations are not always found, for example, between HF-HRV responses and tobacco, alcohol or cannabis use (Evans et al., 2013), and HR reactivity and risky substance use (Evans et al., 2016). Regarding SS reactivity and smoking, one study found associations between lower reactivity and higher risks of tobacco use (Evans et al., 2013), whilst others found no such associations (Evans et al., 2012; Rahal et al., 2023b). In one study, girls, but not boys, who exhibited lower SS reactivity had higher risks of using cannabis or alcohol (Rahal et al., 2023b), whereas in others, SS reactivity was not related to drinking (Evans et al., 2012, 2013) or cannabis use (Evans et al., 2013). These mixed findings highlight the complexity of the relationship between stress reactivity and substance use in adolescents, emphasizing the need for further longitudinal studies to clarify these associations and identify possible differences between measures of stress reactivity.

Notably, there is a dearth of studies that assess whether differences in subjective and neuroendocrine stress reactivity act as explanatory mechanisms in associations between traumatic events and trajectories of substance use in adolescence and young adulthood. In the current study, we contribute to the literature by assessing whether exposure to traumatic events (before the age of 16) predicts stress reactivity around the age of 16, and whether stress reactivity in turn, predicts trajectories of smoking, alcohol, and cannabis use between around 16 and 22 years. To this end, we incorporated a wide range of neuroendocrine measures, which capture acute stress responses of the ANS (HR, PEP, and HF-HRV reactivity) and the HPA-axis (cortisol reactivity), as well as SS reactivity. Furthermore, whilst most previous research typically has assessed substance use at a single time point, we used a longitudinal and person-centred approach in modelling trajectories of substance use, allowing for a distinction between different types of users.

2. Methods

2.1. Study population

We used data from wave 3–5 of the TRacking Adolescents' Individual Lives Survey (TRAILS), a population-based prospective cohort study of Dutch adolescents. A detailed description of the cohort can be found elsewhere (Oldehinkel et al., 2015). At the beginning of the study, 135 schools in the province of Groningen were invited, of which 122 agreed to participate. Adolescents were followed between 2000 and 2013 with assessments around age 11, 14, 16, 19, and 22 years. Around age 16 (wave 3), 744 TRAILS participants were invited to participate in an experimental session, of whom 715 (96.1 %) agreed to take part. Participants with increased risk of mental health problems were over-sampled for this experimental session. Increased risk was determined based on temperament in childhood (i.e., low effortful control, high frustration, and fearfulness), the presence of parental psychopathology, and living in a single parent family. 66.0 % of participants were selected based on having at least one of these risk factors, while 34 % were randomly selected from the remaining participants of the total sample (Bouma et al., 2009). Ethical approval for TRAILS was obtained from the

Dutch national ethics committee Central Committee on Research Involving Human Subjects (#NL38237.042.11). Written informed consent was obtained from both adolescents and their parents prior to inclusion in the study.

2.2. Stress reactivity

2.2.1. The Groningen Social Stress Test (GSST)

The experimental session began with an adaptation phase of 40 min, which was also used to place electrodes, and to administer multiple questionnaires. During the last five min of this phase, baseline cardiovascular measures were taken while participants were seated upright. After completing several other tasks, the Groningen Social Stress Test (GSST) was conducted (Nederhof et al., 2015). The GSST is a procedure modelled after the Trier Social Stress Test, a well-established method for inducing social stress and triggering neuroendocrine stress responses in a controlled laboratory environment. Participants were asked to deliver a 6-min speech about themselves in front of a video camera while seated. Participants were told their speech would be evaluated on delivery, posture, and content, under the neutral observation of a research assistant. Afterward, they were instructed to sit silently for three min due to a technical issue. Subsequently, participants were tasked with repeatedly subtracting 17 from 13,278 for another six min while receiving consistent negative feedback. The feedback included remarks like "You are too slow, try to be faster, we're running out of time," "No, that's incorrect again, start from 13,278," and "Stop moving your hands." (Bouma et al., 2009; Heleniak et al., 2016). The GSST concluded with an additional three min of maintaining a motionless and silent state.

2.2.2. Autonomic nervous system (ANS) reactivity

ANS activity was recorded continuously during the GSST by means of a three-lead electrocardiogram (ECG) and a four-lead impedance cardiogram (ICG) using 3 M/RedDot – Ag/AgCl electrodes (type 2255, 3 M Health Care, Neuss, Germany). The signals were amplified and filtered before digitization at 250 samples/second using a BIOPAC Amplifier-System (MP100). The PreCARSPAN software was used for validating the ECG signal, correcting for artifacts, detecting R-peaks, and for calculating the interbeat intervals (IBI) in ms. Blocks were marked as unusable if they exhibited artifacts lasting longer than 5 s, if the total artifact duration exceeded 10 % of the recording, or if the block length was less than 100 s (Nederhof et al., 2015).

High-frequency heart-rate variability (HF-HRV) was defined as the heart rate variability (HRV) within the high-frequency (HF) band (0.15–0.40 Hz). HF-HRV was quantified by power spectral analysis in the CARSPAN software (Mulder et al., 1988), using estimation techniques based on Fourier transformations of IBI series (Robbe et al., 1987). Measurements that were ≥ 5 standard deviations (SD) above the mean were recoded to missing. Subsequently HF-HRV scores were log transformed (Nederhof et al., 2015). Heart rate (HR) (in heartbeats per min) was calculated by dividing 60.000 by the IBI (Nederhof et al., 2015). PEP is a measure that represents the time interval between the beginning of the electromechanical systole (Q-wave onset) in the ECG and the opening of the aortic valves, which aligns with the B-point in the ICG. To assess thoracic impedance, a BIOPAC NICO100C Non-invasive Cardiac Output Module was used. Manual scoring of the B-points was conducted by a highly trained evaluator using the VU-AMS interactive software. In cases where there was uncertainty about the B-point, a second rater was consulted for discussion. Outliers were examined, and the quality of PEP scores was subjectively rated on a scale from 0 to 10. PEP data were deemed unreliable if the quality of the PEP measurement was low (i.e., score < 6) or if the signal contained too many artifacts, including cases where participants had arrhythmias or extra systoles (Nederhof et al., 2015).

To assess HR, HF-HRV, and PEP reactivity, we regressed the average of, respectively the mean HR, HF-HRV, and PEP during the first three min of the speech and math task on the baseline measurement of each of

the three measures, which was taken during the last five min of the adaptation phase. We then saved the standardized residuals. This approach is consistent with the previous literature (c.f., Heleniak et al., 2016) and was chosen as ANS responses tend to be greatest shortly after the beginning of each task. Higher HR reactivity scores are consistent with steeper increases in HR, indicative of a stronger response of both the SNS and PNS (Nederhof et al., 2015). On their original scales, lower PEP reactivity scores are indicative of a stronger response of the SNS, and lower HF-HRV reactivity scores are indicative of a stronger response of the PNS (Nederhof et al., 2015). For similar interpretation, such that higher scores indicate a stronger stress response, we multiplied PEP and HF-HRV reactivity scores with -1 . Valid HR, PEP, and HF-HRV-reactivity scores were available for, respectively, $N = 651$, $N = 506$, and $N = 644$ participants.

2.2.3. Cortisol reactivity

Cortisol levels were assessed using the Salivette sampling device (Sarstedt, Nümbrecht, Germany), which contains a small swab participants had to chew for 60 s. Measurements were taken in duplicate using an in-house radioimmunoassay (RIA) applying a polyclonal rabbit cortisol antibody and 1,2,6,7 ^3H cortisol (Amersham) as tracer (Bosch et al., 2012). The intra-assay coefficient of variation was 8.2 % for 1.5 nM concentrations, 4.1 % for 15 nM concentrations, and 5.4 % for 30 nM concentrations. The inter-assay coefficients of variation were 12.6 %, 5.6 %, and 6.0 %, respectively. The lower limit of detection was 0.9 nM. Cortisol levels were assessed just before the start of the GSST, and directly after the GSST. Given that salivary cortisol is only detectable approximately 20 min after release by the adrenal glands, this latter measurement reflects cortisol levels during the speech task (Bouma et al., 2009). We set values deviating more than 5 SD from the mean to missing. To calculate cortisol reactivity, we regressed post-GSST cortisol levels on pre-GSST levels, and then saved the standardized residuals. Cortisol reactivity was set to missing for girls using oral contraception ($N = 132$), as well as participants using medications known to influence cortisol levels (e.g., corticosteroids, SSRIs) ($N = 34$) (Bouma et al., 2009). Cortisol reactivity was available for $N = 537$ participants.

2.2.4. Subjective stress (SS) reactivity

The Self-Assessment Manikin, a non-verbal pictorial method, was employed to evaluate SS levels both during the adaptation phase and right after the GSST. This technique gauges an individual's psychological response to stressors by selecting one of nine ordered pictures to express the subjective intensity of arousal, unpleasantness, and dominance (i.e., feeling of control) (Bradley and Lang, 1994). The responses were recoded into three nine-point scales (ranging from 1 to 9), where higher scores indicated higher levels of arousal, unpleasantness, and lower levels of control (i.e., low dominance), of which the mean was taken as indicator of SS. SS reactivity was calculated by regressing SS right after the GSST on SS at baseline and then exporting the standardized residuals. SS reactivity was available for $N = 709$ participants.

2.3. Traumatic experiences

Traumatic experiences prior to the age of 16 were assessed retrospectively around age 22 (wave 4) by self-report. Experiences of sexual abuse were assessed using five items (Cronbach's $\alpha = 0.90$). Five additional items were used to assess other traumatic experiences, which included natural disasters, life-threatening accidents, being victim of physical violence or a terror attack, witnessing severe injury or death of another person, and being threatened with a weapon, held captive or abducted (Cronbach's $\alpha = 0.57$). Response options for both scales were 1) 'no, never', 2) 'yes, once', and 3) 'yes, more than once'. These questions were developed by TRAILS (Heleniak et al., 2016). Physical abuse by parents was assessed using six items of a modified version of the Parent-Child Conflict Tactics Scales (Cronbach's $\alpha = 0.72$), a validated self-report questionnaire (Straus et al., 1998). Example items are

“My mother and/or father hit me with a belt, brush, stick or another hard object”, and “My mother and/or father threatened me with a knife or another weapon”. The response options were 1) ‘never’, 2) ‘1 or 2 times’, 3) ‘sometimes’, 4) ‘often’, and 5) ‘very often’. Mean scores were calculated for each of the three scales separately (i.e., sexual abuse, other traumatic events, physical abuse) and then z-score transformed. Higher scores on each of the three scales indicate either exposure to a greater variety of traumatic event types or repeated exposure to a limited number of events. Detailed information on the frequency of each traumatic event is given in Fig. 1 and Table S4. Lastly, consistent with previous research from the TRAILS study (Heleniak et al., 2016), we log-transformed and standardized each of the three scales before creating a summed trauma exposure score, which assigns equal weight to each of the domains (sexual abuse, physical abuse, and other traumatic events).

2.4. Substance use

2.4.1. Smoking

Adolescents and young adults were surveyed about their recent smoking behaviours over the preceding four weeks from wave 3–5. Their responses were transformed to estimate the average number of cigarettes smoked per day: 0 (non-smokers), 1 (fewer than one cigarette daily), 3 (1–5 cigarettes daily), 8 (6–10 cigarettes daily), 15 (11–20 cigarettes daily), 20 (> 20 cigarettes daily).

2.4.2. Alcohol use

Alcohol use from wave 3–5 was evaluated by self-report utilizing a quantity-frequency measure. Quantity-frequency measures for alcohol

use have demonstrated adequate/good validity and reliability across studies (McKenna et al., 2018). The assessment of frequency involved inquiring about regular alcohol drinking days during the week (Monday to Thursday) and weekend (Friday to Sunday) separately. Quantity was assessed by the average number of alcoholic beverages consumed on a typical weekday or weekend day, using a 9-point scale ranging from 1 (‘I never drink on a weekday/weekend day’) to 9 (‘11 glasses or more.’) The quantity scores for weekdays and weekend days were multiplied by the corresponding frequency scores. Then both scores were summed to calculate a quantity-frequency measure.

2.4.3. Cannabis use

Recent cannabis use was assessed from wave 3–5 by self-report by enquiring about the number of times marijuana (weed) or hashish was consumed in the previous four weeks. Answers were recoded to 0 (no use), 1 (one time), 2 (two times), 3 (three times), 4 (four times), 5 (five times), 6 (six times), 7 (seven times), 8 (eight times), 9 (nine times), 10 (ten times), 15 (11–19 times), 29 (20–39 times), and 40 (≥ 40 times).

2.5. Covariates

Previous research has identified pubertal stage and adiposity as shared predictors of both adolescent neuroendocrine functioning and substance use (e.g., Castellanos-Ryan et al. 2013; Gearhardt et al. 2018; Hillman et al. 2012; Romeo 2010), potentially confounding stress reactivity-substance use associations. Therefore, we included these variables as covariates in our primary analyses. However, as traumatic experiences may influence pubertal timing and adiposity (Colich and McLaughlin, 2022; Schroeder et al., 2021), controlling for these factors

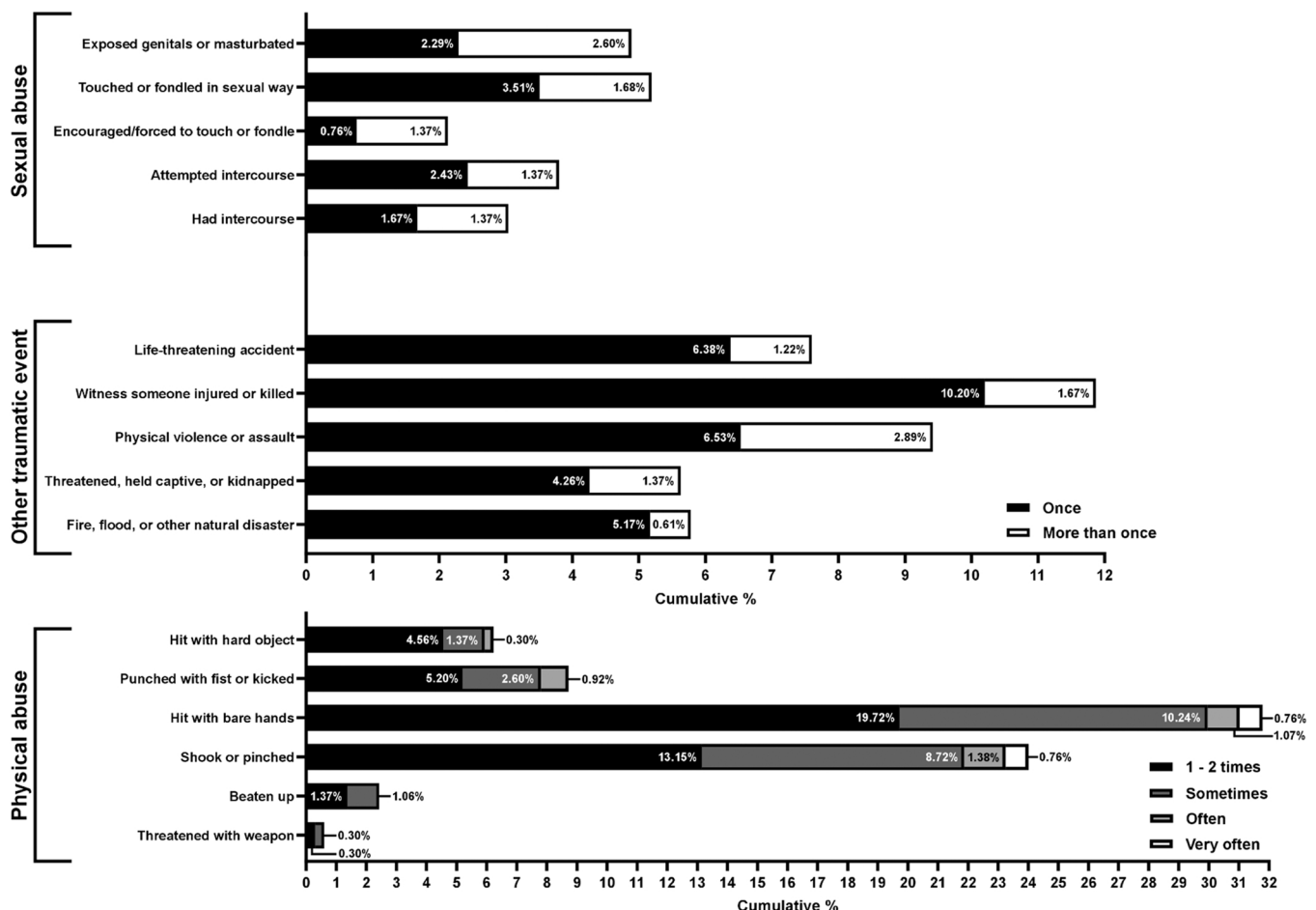


Fig. 1. Descriptive statistics of the items used to assess traumatic experiences.

could represent overadjustment. This may lead to an underestimation of associations of traumatic experiences with stress reactivity, and, by extension, associations with substance use. Therefore, we conducted sensitivity analyses excluding adiposity and pubertal stage, which yielded results consistent with our main findings (Figures S1-S3). Pubertal stage was assessed at wave 3 using the Physical Development Scale (PDS), a validated scale (Schmitz et al., 2004) which evaluates the progression of five traits for each sex (i.e., breast development and menarche in girls, voice change and facial hair growth in boys, height growth, skin changes, and body hair for both sexes). The PDS yielded a Cronbach's alpha of 0.73 for boys and 0.52 for girls. Adiposity was operationalized using *body fat percentage*, measured via hand-to-foot bioelectrical impedance analysis according to the Deurenberg equation using the BIA 101 (Akern®, Pontassieve, Italy) at the GSST date.

Sociodemographic covariates were selected a priori based on past research on traumatic experiences, neuroendocrine stress reactivity, and substance use (e.g., Evans et al. 2013; Hines et al. 2023; Janssens et al. 2016) and included age at the date of the GSST, sex, parental socioeconomic status (SES) measured at wave 1 (around age 11), and ethnicity. Adolescents were categorized as having non-Dutch ethnicity if at least one of their parents was born outside the Netherlands. SES was determined by taking the mean of five indicators (standardized): maternal and paternal educational attainment, maternal and paternal occupational position (based on the International Standard Classification of Occupations), and family income.

2.6. Analytic approach

2.6.1. Calculation of trajectories

We identified growth trajectories of drinking behaviour, smoking, and cannabis use from around age 16–22 using latent class growth analyses (LCGA) with linear slopes, as recommended for models with three measurement occasions (Smith and Dunn, 2023). Zero-inflated Poisson (ZIP) models were used to accommodate zero-inflation and overdispersion of the three substance use outcomes. The ideal number of classes was established through several criteria: (a) an improvement in model fit, as evidenced by decreases in the Akaike Information Criterion (AIC), the Bayesian Information Criterion (BIC), and the Sample-Size adjusted Bayesian Information Criterion (SABIC); (b) achieving a satisfactory level of classification, as indicated by an entropy value > 0.80; (c) a statistically significant improvement in fit, as indicated by the Bootstrap Likelihood Ratio Test (BLRT) and the Lo-Mendell-Rubin likelihood ratio test (LMR-LRT), (d) an adequate sample size for each class (> 2 %), and (e) theoretical considerations (Peeters et al., 2019) (Table S2). To account for classification uncertainty when trajectories are used in further analyses, the three-step VAM approach was used (Asparouhov and Muthén, 2014; McLarnon and O'Neill, 2018). In this approach, first LCGMs are estimated without predictors, whilst full information maximum likelihood (FIML) is used to address missing observations on substance use. Then the most likely trajectory memberships are exported for each participant. In a third step, the mixture model is re-estimated with all predictors of trajectories, whilst fixing measurement parameters at values that account for the classification uncertainty in the LCGA obtained in step 1. Missing data in this final estimation step were addressed using multiple imputations by chained equations with the mice package (version 3.17.0) in R 4.5.0, which generated 100 imputed datasets with 100 between-imputation iterations (Asparouhov and Muthén, 2021; McLarnon and O'Neill, 2018; van Buuren and Groothuis-Oudshoorn, 2011). The amount of missing information per variable can be found in Table S3. We additionally conducted sensitivity analyses of all models, in which we did not adjust for classification uncertainty (Figure S4-S6). The results were mostly consistent with our main analysis, suggesting that the impact of classification uncertainty on our models is minimal.

2.6.2. Structural equation model (SEM)

We conducted structural equation models (SEM) to evaluate 1) associations between exposure to traumatic events and stress reactivity measures, 2) associations between stress reactivity measures and the substance use trajectories identified using LCGA, and 3) associations between exposure to traumatic events and substance use trajectories. Separate models were run for each stress reactivity measure (i.e., Cortisol, PEP, HF-HRV, HR, SS reactivity), and each substance use outcome (i.e., smoking, alcohol use, and cannabis use). Since no substantial non-linearities in associations were detected (Figures S7-S13), linear regression was used to predict stress reactivity measures, and multinomial logistic regression to model substance use trajectories. All regressions were adjusted for all covariates (i.e., age, ethnicity, sex, parental SES, pubertal stage, and body fat percentage), and all continuous predictors z-score transformed to facilitate interpretation. We used the robust maximum likelihood estimator (MLR) to adjust for non-normality. Analyses were conducted in Mplus 8.11.

3. Results

3.1. Descriptive statistics

Key descriptive statistics of the study sample are displayed in Table 1. Fig. 1 and Table S4 provide detailed information on adolescents' traumatic experiences by visualizing responses to each of the corresponding items. The most common forms of physical abuse by parents involved being struck with bare hands (31.80 %) or being shaken or pinched (24.01 %), with these percentages reflecting participants who reported at least one occurrence. In contrast, more severe forms of physical abuse, such as being punched with a fist or kicked hard (8.72 %), hit with a hard object (6.23 %), or beaten up (2.43 %), were reported far less frequently. Regular physical abuse was rare in the sample, with only a small proportion of participants indicating that such acts occurred 'often' or 'very often.' All items measuring physical abuse

Table 1
Characteristics of the study population.

	N = 715	
Traumatic experiences, mean (SD)	-0.01 (1.96)	
Stress measures, mean (SD)	Baseline	GSST
Cortisol	3.43 (2.05)	4.69 (2.93)
HR	75.62 (11.14)	80.87 (12.08)
HF-HRV	6.69 (1.21)	6.98 (1.00)
PEP	124.59 (19.34)	111.03 (19.03)
SS	3.02 (1.00)	4.54 (1.53)
Smoking trajectories, N (%)		
No use	427 (59.97)	
Moderate increasing	114 (16.01)	
Heavy increasing	171 (24.02)	
Alcohol use trajectories, N (%)		
Stable low	385 (54.30)	
Moderate increasing	222 (31.31)	
Decreasing	48 (6.77)	
Heavy increasing	54 (7.62)	
Cannabis use trajectories, N (%)		
No use	530 (74.54)	
Stable low	88 (12.38)	
Decreasing	37 (5.20)	
Late increase	24 (3.38)	
Early increase	32 (4.50)	
Covariates		
Age at GSST, mean (SD)	15.61 (0.66)	
Non-Dutch ethnicity, N (%)	66 (9.23)	
Male sex, N (%)	351 (49.09)	
Parental SES, mean (SD)	0.07 (0.76)	
Pubertal stage, mean (SD)	-0.02 (1.01)	
Body fat percentage, mean (SD)	28.34 (5.62)	

SD = standard deviation; GSST = Groningen Social Stress Test; HR = heart rate; HF-HRV = high-frequency heart rate variability; PEP = pre-ejection period; SS = subjective stress; SES = socioeconomic status.

were significantly correlated with one another (Table S5), with most correlations exhibiting medium ($r > 0.30$) to large ($r > 0.50$) effect sizes (Cohen, 1988). Sexual abuse was relatively uncommon in the sample. The most frequently reported experiences were unwanted touching or fondling (5.19 %) and exposure to an adult's genitals or masturbation (4.89 %). Reports of attempted (3.79 %) or completed sexual intercourse (3.03 %) were even less common. We found strong correlations (i.e., > 0.50) among all items assessing sexual abuse, indicating frequent co-occurrence of different types of sexual abuse. Approximately half of the adolescents who reported any sexual abuse reported multiple incidents of sexual abuse. Regarding other traumatic events, the most frequently reported experiences were witnessing someone being seriously injured or killed (11.87 %) and experiencing physical violence or an assault (9.42 %). Less common were exposure to fires, floods, or natural disasters (5.78 %), and being threatened with a weapon, held captive, or kidnapped (5.62 %). Most participants who experienced these events reported only one occurrence; however, physical violence or assault was an exception, with slightly over a quarter of affected participants experiencing multiple exposures. We observed small (i.e., > 0.10) to medium-sized (i.e., > 0.30) correlations among some of the items, such as involvement in a life-threatening accident and witnessing severe injury or death ($r = 0.35$).

3.2. Trajectories of smoking, cannabis use, and alcohol use

For smoking, our LCGA identified three trajectories (Table 2, Fig. 2): The heavy increasing trajectory (24.02 %) was characterized by high levels of smoking already around age 16, followed by a further increase until around age 19, and persistently high levels until around age 22. Adolescents following the moderate increasing trajectory (16.01 %) showed lower levels of smoking around age 16, which increased slightly until around age 19, and then further increased until around age 22. The third trajectory (no use) mainly consisted of non-smokers (59.97 %). Regarding alcohol use, we identified four trajectories: Both the heavy increasing (7.62 %) and decreasing trajectories (6.77 %) were characterized by an early initiation and escalation of alcohol use until age 16. However, whilst adolescents following the decreasing trajectory progressively moderated their drinking behaviour until around age 22, the alcohol use of adolescents following the heavy increasing trajectory escalated further until age 19, and then remained very high until around age 22. Adolescents following the moderate increasing trajectory (31.31 %) used less alcohol around age 16 than the aforementioned two trajectories. Subsequently, their alcohol use showed gradual increases

until about age 19, continuing to rise slightly until around age 22. The stable low trajectory was characterized by persistently low levels of drinking (54.30 %). For cannabis use, we found five trajectories: The early increasing trajectory (4.50 %) was characterized by high levels of cannabis use around age 16 followed by a further escalation until around age 19, and a subsequent decrease in use until around age 22. The decreasing trajectory (5.20 %) was also defined by an early escalation of cannabis use, but, nevertheless, lower levels of use around age 16 than the heavy increasing trajectory. Adolescents following this trajectory then progressively lowered their use until around age 22. The late increasing trajectory (3.38 %) was characterized by very low levels of use around age 16, followed by a moderate increase until age 19, and then a steep escalation up to around age 22. The stable low trajectory was characterized by consistently low levels of use (12.38 %), whilst the no use trajectory consisted mainly of abstainers (74.54 %).

3.3. Associations between traumatic experiences and substance use trajectories

Regarding smoking, adolescents with more traumatic experiences had significantly higher odds of following the heavy increasing trajectory (vs. no use) (Fig. 3). For alcohol use (Fig. 4), traumatic experiences were associated with higher odds of following the heavy increasing trajectory, and to a lesser extent the decreasing trajectory (vs. stable low). Considering cannabis use, adolescents who reported more traumatic experiences had higher odds of following the early increasing, decreasing and late increasing trajectories, and to a lesser extent the stable low trajectory (vs. no use) (Fig. 5). Overall, traumatic experiences were associated with following the more problematic trajectories of tobacco, alcohol, and cannabis use, characterized by an early onset and escalation of use over time, and to a lesser extent with a trajectory marked by consistently low levels of cannabis use.

3.4. Associations between traumatic experiences and stress reactivity

We found no significant associations between traumatic experiences and any of the neuroendocrine and subjective stress reactivity measures (Figs. 3–5).

3.5. Associations between stress reactivity and substance use trajectories

Regarding smoking (Fig. 3), adolescents with blunted neuroendocrine stress reactivity, in terms of HR, HF-HRV, PEP, and cortisol

Table 2
Descriptive statistics of substance use trajectories.

Comparative analysis of substance use trajectories												
	Smoking			Alcohol use		Cannabis use						
	No use	Moderate increasing	Heavy increasing	Stable low	Moderate increasing	Decreasing	Heavy increasing	No use	Stable low	Decreasing	Late increase	Early increase
N (%)	427 (59.97)	114 (16.01)	171 (24.02)	385 (54.30)	222 (31.31)	48 (6.77)	54 (7.62)	530 (74.54)	88 (12.38)	37 (5.20)	24 (3.38)	32 (4.50)
	Mean (SD)	Mean (SD)	Mean (SD)	Mean (SD)	Mean (SD)	Mean (SD)	Mean (SD)	Mean (SD)	Mean (SD)	Mean (SD)	Mean (SD)	Mean (SD)
Age around 16 (wave 3)	0.03 (0.17)	1.69 (2.65)	8.67 (7.26)	2.59 (2.87)	6.98 (4.91)	29.51 (18.20)	17.00 (9.51)	0.03 (0.16)	1.70 (2.24)	8.19 (11.98)	1.79 (2.59)	15.17 (16.70)
Age around 19 (wave 4)	0.08 (0.27)	2.80 (3.10)	11.63 (6.61)	4.18 (3.26)	12.03 (5.95)	19.26 (11.56)	32.78 (15.28)	0.03 (0.18)	1.41 (1.56)	6.81 (7.62)	8.26 (6.55)	23.85 (17.68)
Age around 22 (wave 5)	0.12 (0.33)	4.19 (3.65)	11.15 (7.05)	4.53 (3.40)	14.53 (7.37)	10.86 (5.95)	31.73 (15.11)	0.03 (0.16)	1.14 (1.43)	4.50 (3.30)	21.88 (10.12)	18.04 (16.87)

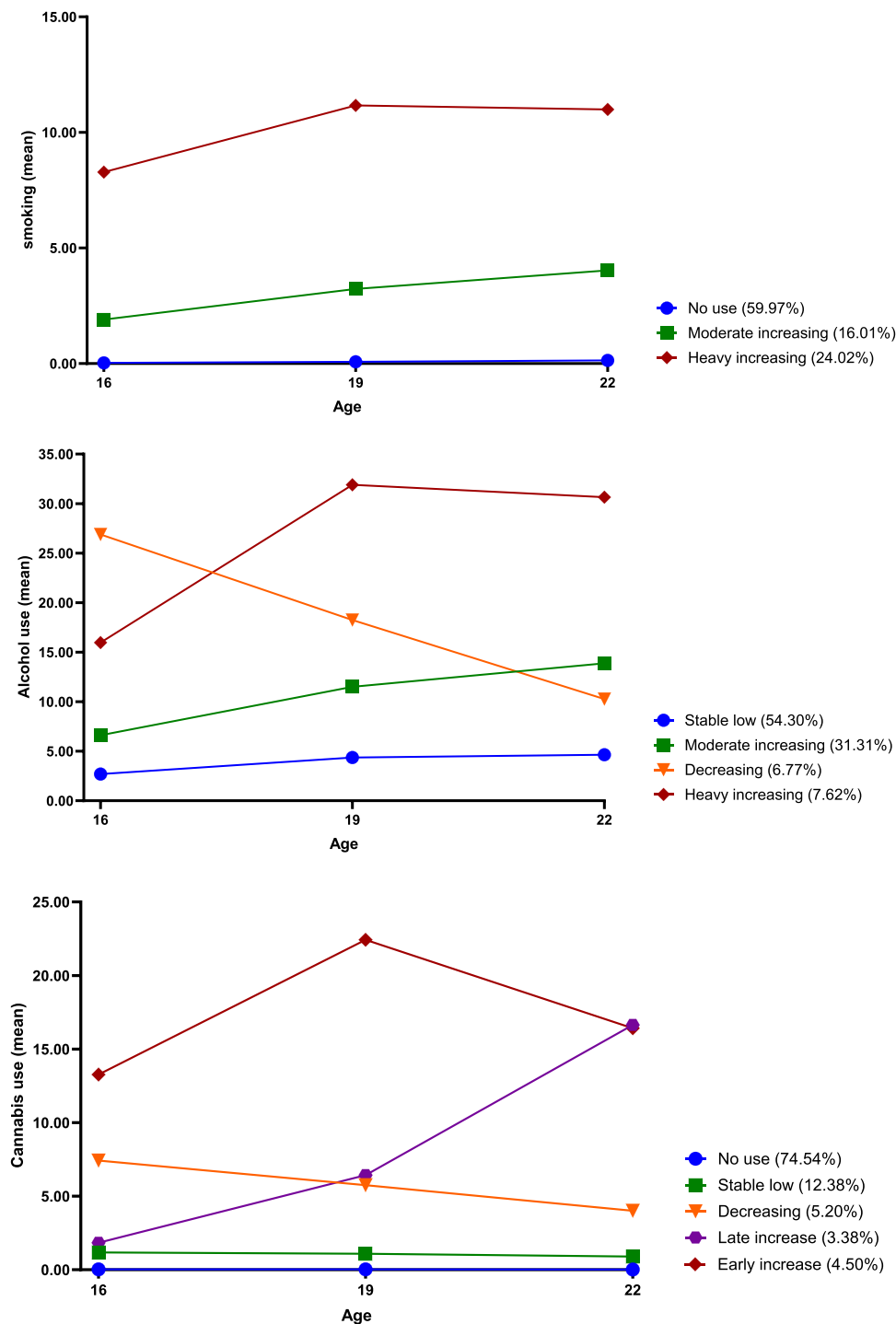


Fig. 2. Trajectories of substance use from around age 16 to around age 22.

reactivity, had higher odds of following the heavy increasing trajectory (vs. no use). SS reactivity was not associated with smoking. For alcohol use (Fig. 4), adolescents with blunted HF-HRV reactivity had higher odds of following the heavy increasing trajectory (vs. stable low). Adolescents with blunted HR reactivity had higher odds of following the decreasing and moderate increasing trajectories (vs. stable low). We found no significant associations of cortisol, PEP, and SS reactivity with alcohol use trajectories. Regarding cannabis use (Fig. 5), adolescents with blunted cortisol reactivity had higher odds of following the early increasing trajectory (vs. no use). In summary, blunted neuroendocrine stress reactivity was inconsistently linked to early increasing and/or

more intensive substance use trajectories.

4. Discussion

Adolescents exposed to traumatic events, including physical and sexual abuse, are at higher risk of developing problematic substance use behaviours. Yet, the mechanisms underlying these effects are insufficiently understood. This study aimed to address this gap by examining acute subjective and neuroendocrine stress reactivity – specifically cortisol, heart rate (HR), high-frequency heart rate variability (HR-HRV), and pre-ejection period (PEP) reactivity – as possible pathways. In

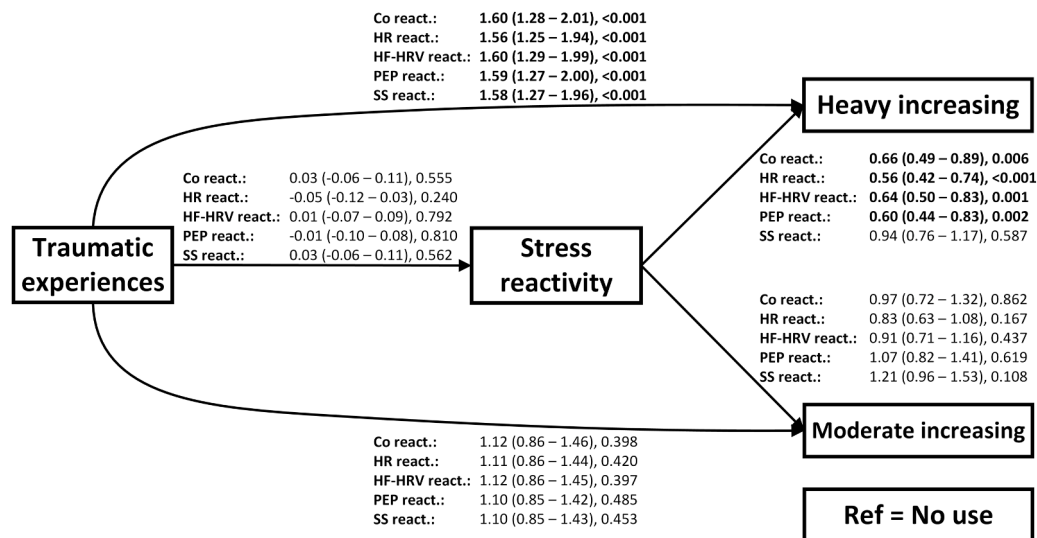


Fig. 3. Associations between traumatic experiences until around age 16, stress reactivity measures around age 16, and smoking trajectories from around age 16 to around age 22. Adjusted for covariates, which were allowed to predict all variables in the model: age, ethnicity, sex, parental SES, pubertal stage, and body fat percentage. Linear regression (standardized beta-coefficient (95 % confidence interval)) was used to predict stress reactivity measures, and multinomial logistic regression (odds ratio (95 % confidence interval)) to predict substance use trajectories. **Boldface** denotes statistical significance at $p < 0.05$.

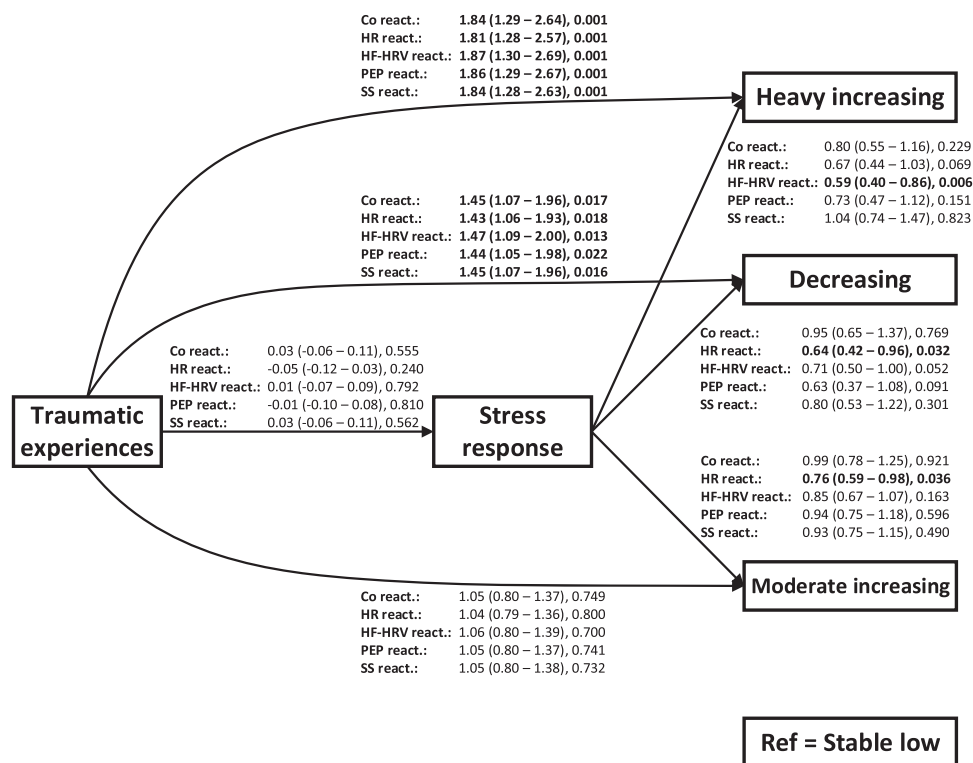


Fig. 4. Associations between traumatic experiences until around age 16, stress reactivity measures around age 16, and alcohol use trajectories from around age 16 to around age 22. Adjusted for covariates, which were allowed to predict all variables in the model: age, ethnicity, sex, parental SES, pubertal stage, and body fat percentage. Linear regression (standardized beta-coefficient (95 % confidence interval)) was used to predict stress reactivity measures, and multinomial logistic regression (odds ratio (95 % confidence interval)) to predict substance use trajectories. **Boldface** denotes statistical significance at $p < 0.05$.

line with past research, traumatic experiences before age 16 consistently predicted more severe and problematic patterns of substance use during adolescence and young adulthood. Additionally, we observed inconsistent associations between neuroendocrine stress hyporeactivity and trajectories of higher substance use, which differed depending on the neuroendocrine measure and the type of substance under consideration. No associations between subjective stress reactivity and trajectories of

substance use were found. Unexpectedly, acute stress reactivity did not explain associations between traumatic experiences and substance use trajectories, as we detected no associations between traumatic experiences and acute stress reactivity measures. This may point to other unassessed mechanisms linking traumatic experiences to substance use trajectories.

Our results corroborate past research showing robust associations

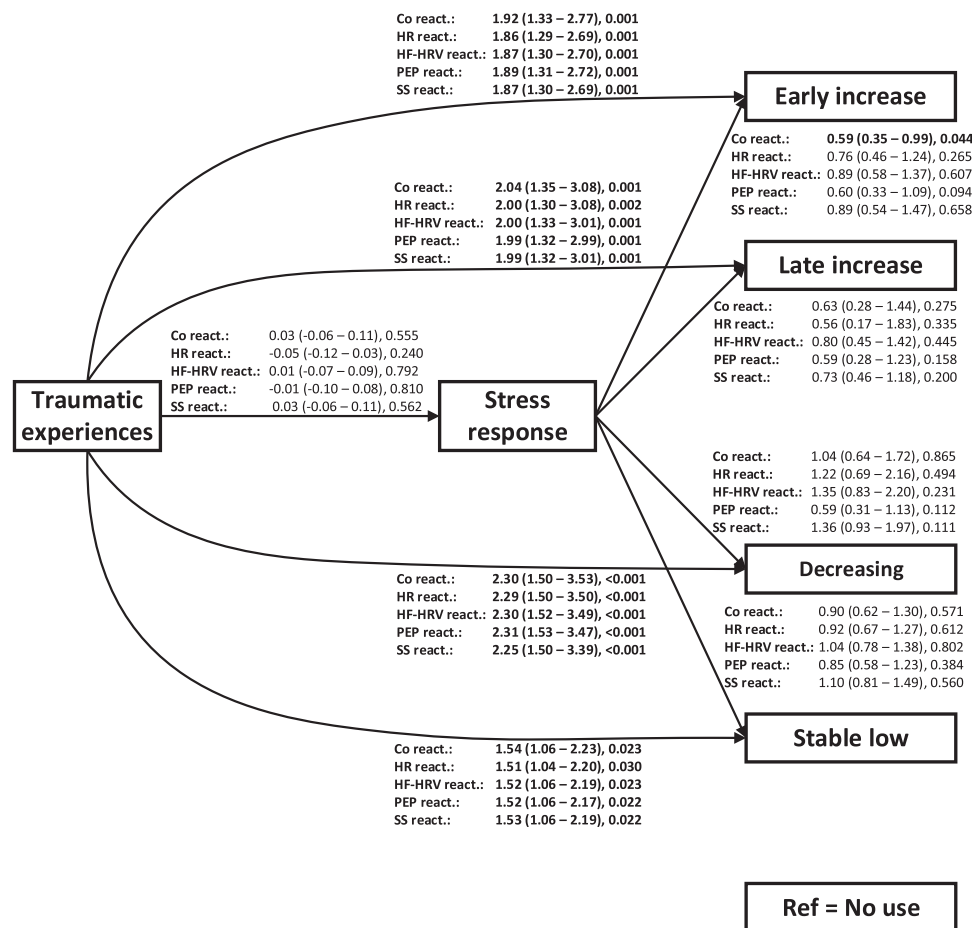


Fig. 5. Associations between traumatic experiences until around age 16, stress reactivity measures around age 16, and cannabis use trajectories from around age 16 to around age 22. Adjusted for covariates, which were allowed to predict all variables in the model: age, ethnicity, sex, parental SES, pubertal stage, and body fat percentage. Linear regression (standardized beta-coefficient (95 % confidence interval)) was used to predict stress reactivity measures, and multinomial logistic regression (odds ratio (95 % confidence interval)) to predict substance use trajectories. **Boldface** denotes statistical significance at $p < 0.05$.

between traumatic experiences early in life and problematic substance use trajectories characterized by an early initiation and subsequent heavy escalation of use (Kirsch and Lippard, 2022). For smoking, traumatic experiences predicted an early and strong increase in tobacco use throughout adolescence and young adulthood (vs. no use). Also regarding alcohol use, we found significant associations between traumatic experiences and trajectories defined by an early escalation of use: Traumatic experiences predicted an early initiation followed by a strong increase in drinking, as well as an early initiation followed by a steady decline over the subsequent years (vs. consistently low alcohol use). Furthermore, traumatic experiences were similarly associated with three trajectories characterized by higher levels of cannabis use over the course of adolescence and young adulthood (vs. no use). Whilst we found no associations between traumatic experiences and the moderate increasing trajectories of smoking and alcohol use, traumatic experiences were additionally associated with a cannabis use trajectory characterized by consistently relatively low levels of use (vs. no use). Our findings are similar to a previous study, which found associations between physical abuse and both early and late onset regular cannabis use, as well as early persistent occasional use (vs. low or no use) (Hines et al., 2023). Occasional cannabis use in adolescence, perhaps more so than occasional alcohol use or smoking, could be considered a marker of risk and, accordingly, has been linked to a multitude of adverse outcomes, including lifetime suicide attempts, violent behaviour, and later substance use-related problems (Degenhardt et al., 2010; Huas et al., 2008). Our findings are critical, as an early exposure to addictive substances, including cannabis, alcohol, and tobacco heightens the risk of

developing SUDs later in life (Kirsch and Lippard, 2022). In that way, early-onset adolescent substance use may serve as intermediate step linking traumatic experiences to later SUDs (Kirsch and Lippard, 2022).

In contrast to our expectations, associations between traumatic experiences and substance use trajectories were not explained by differences in neuroendocrine or psychological stress reactivity, as associations between traumatic experiences and stress reactivity measures were absent. This points to mechanisms we have not evaluated. Trauma survivors often face greater challenges in regulating emotions, which can lead to heightened impulsivity (Oshri et al., 2018; Weiss et al., 2023) and a tendency toward negative emotion-focused or disengagement coping strategies to escape or avoid stressors (Milojevic et al., 2018; Vaughn-Coaxum et al., 2018). Adolescents prone to these coping styles may be more likely to turn to substances as a way to numb difficult emotions (McConnell et al., 2014), a behavior that impulsivity may further reinforce (Oshri et al., 2018). These mechanisms, alongside associated factors, including mental health problems such as PTSD symptoms, may explain the increased risks of following problematic substance use trajectories amongst trauma-exposed adolescents and young adults (Kirsch and Lippard, 2022; McConnell et al., 2014).

Our findings support the hypothesis that blunted neuroendocrine reactivity serves as a biomarker of substance use risk, independently predicting substance use trajectories regardless of traumatic experiences. Blunted reactivity of all neuroendocrine measures was consistently associated with following the heavy increasing (vs. no use) smoking trajectory. We found inconsistent associations between blunted ANS reactivity and following the three trajectories characterized by

higher levels of alcohol use (vs. consistently low alcohol use). Regarding cannabis use, we only found one significant association between blunted cortisol reactivity and following the early increasing trajectory (vs. no use). Our findings are largely in line with previous research that mainly found associations between neuroendocrine hyporeactivity and increased substance use in adolescents (Evans et al., 2012; Marceau et al., 2022; Rahal et al., 2023a). Adolescents with hyporeactive stress physiology may have lower levels of fear and seek out experiences that increase arousal to deal with chronic under-arousal, such as substance use (Chaplin et al., 2018; Marceau and Abel, 2018). Over time, chronic substance use can lead to further alterations in stress neuroendocrine systems and even greater reductions in acute neuroendocrine stress reactivity, which may further reinforce addictive behaviour. Accordingly, chronic administration of THC, alcohol, and nicotine led to alterations in HPA-axis functioning in animal studies (Fosnocht and Briand, 2016). However, it is unlikely that such reverse-causal mechanisms can fully explain associations between blunted neuroendocrine reactivity and substance use. This line of reasoning is supported by studies that have demonstrated that blunted neuroendocrine reactivity precedes the onset of substance use in at-risk individuals. Several studies have found blunted cortisol reactivity already in prepubertal children of substance abusing fathers (Dawes et al., 1999; Hardie et al., 2002; Moss et al., 1995), and this result was not explained by differences in family dysfunction (Hardie et al., 2002). Furthermore, blunted neuroendocrine reactivity has been found in behavioural addictions (e.g., problem gambling, and exercise addictions) (Heaney et al., 2011; Pangborn et al., 2024), suggesting that reduced neuroendocrine stress reactivity may reflect biopsychological processes underlying multiple addictions, including those that do not involve substances. These processes may involve associated risk factors, such as trait impulsivity, high reward sensitivity, and sensation-seeking (Kirsch and Lippard, 2022).

We found no associations between SS reactivity and substance use trajectories. Very few past studies on acute stress reactivity and adolescent substance use have included measures of SS reactivity, alongside neuroendocrine reactivity, yielding mixed results (Chaplin et al., 2018; Evans et al., 2012, 2013; Rahal et al., 2023b). Given that SS reactivity, as assessed in standardized stress tests, often does not correlate well with neuroendocrine reactivity (e.g., in terms of cortisol) (Campbell and Ehlert, 2012), it is of added value for future studies to also include SS reactivity to provide a more comprehensive understanding of acute stress reactivity, capturing both psychological and biological dimensions.

5. Limitations

This study has several strengths, including its relatively large sample, a comprehensive neuroendocrine assessment, and long follow-up over a six-year period, allowing us to use person-centred approaches to identify typologies of substance users. However, this study also has several limitations. First, we cannot rule out reverse causation in associations between neuroendocrine stress reactivity and substance use, as mentioned above. However, adolescents were still relatively young (around 16 years) when the GSST was conducted, suggesting that substance use may have had limited time to influence stress physiology. Second, we assessed traumatic experiences retrospectively, which may have introduced recall bias. Third, our item-level analysis suggests that repeated exposure to severe traumatic events (e.g., repeated sexual abuse) was relatively rare in the sample. Moreover, our measures did not provide detailed information on the frequency of traumatic events, which is essential for assessing trauma chronicity. Because persistent and severe trauma is likely to be more strongly associated with altered neuroendocrine stress reactivity, this combination of a low prevalence of repeated severe trauma and measurement limitations may have contributed to the absence of an observed association between traumatic experiences and stress reactivity. Supporting this notion, Marques-Feixa et al. (2023) found that children and adolescents exposed to more

frequent or severe child maltreatment showed the strongest blunting of cortisol reactivity. Similarly, studies on substance use suggest that frequent and prolonged exposure to traumatic events may have particularly strong associations with adverse outcomes (Gabrielli et al., 2016; Jonson-Reid et al., 2012). Notably, research on the role of trauma chronicity in neuroendocrine stress reactivity remains limited (Schär et al., 2022), leaving an important gap in the literature. Future studies could address this by including more detailed assessments of the timing, frequency, severity, and chronicity of traumatic events. It may also be valuable to focus on high-risk populations, such as institutionalized adolescents, where repeated exposure to trauma is more common than in the TRAILS study. Fourth, our study focused on the most severe forms of traumatic events, which loosely align with DSM-5 Criterion A for posttraumatic stress disorder (PTSD), which includes direct or indirect exposure to actual or threatened death, serious injury, or sexual violence (Krupnik, 2019). Less severe childhood adversities, such as perceived discrimination, may have distinct associations with neuroendocrine functioning (Hosseini-Kamkar et al., 2021), as described above. Fifth, attrition might have influenced the results of our study. Although we implemented FIML and multiple imputations to manage missing data, higher dropout of adolescents with less favourable conditions (e.g. lower parental SES, more traumatic experiences) may still have affected our results. As these characteristics are also important determinants of adverse outcomes in young adulthood, further research on at-risk groups is necessary (Caspi et al., 2016). Sixth, this study only considered acute stress reactivity assessed in the context of a standardized stress test. Other markers, such as diurnal cortisol or cortisol recovery, may be differently related to traumatic experiences and substance use than the measures used in this study. Seventh, even though compared to much previous research on adolescents (e.g., Evans et al. 2016; Evans et al. 2012; Rahal et al. 2023b) our sample is relatively large, insufficient statistical power may have hampered our ability to detect small associations, particularly regarding substance use trajectories not many adolescents followed. The small size of multiple trajectory groups also resulted in insufficient power to assess potential sex differences in associations. Future person-centred research may require larger sample sizes to evaluate associations with smaller, often higher-risk, trajectories, whilst also taking into account potential sex differences. Eighth, the use of data that were already collected precluded us from conducting power analyses prior to data collection. Post-hoc power analyses are generally not recommended in the statistical literature (Heinsberg and Weeks, 2022). Last, the generalizability of findings may be limited, given substantial cross-national and temporal variations in adolescent substance use and related social norms. The Netherlands has historically maintained relatively liberal substance use policies, particularly regarding cannabis use, permitting regulated sales to adults (18+) for many decades (Simons-Morton et al., 2010). Nevertheless, recent years have seen growing recognition of the harms of underage substance use and stricter regulations on alcohol and tobacco sales and advertising (Boderie et al., 2024; de Looze et al., 2017). For example, while legal for adults (18+), smoking has been increasingly discouraged through comprehensive measures, including public smoking bans, higher taxes, graphic warnings, and advertising restrictions (Boderie et al., 2024). In addition, the legal purchasing age for mild alcoholic beverages was raised from 16 to 18, matching the previously existing limit for stronger alcohol (Schelleman-Offermans et al., 2017), and Dutch parents have adopted stricter alcohol-related parenting practices compared to previous generations (de Looze et al., 2017). These shifts have coincided with declining adolescent smoking and drinking, but also cannabis use since the beginning of the century (Bommelé et al., 2023; de Looze et al., 2017; Rombouts et al., 2023), potentially reflecting a broader societal shift toward discouraging underage substance use. However, the TRAILS cohort came of age under earlier, more permissive norms, particularly regarding alcohol use. During their youth, the Netherlands had one of the highest rates of adolescent (12–16 years) alcohol consumption internationally, and 16-year-olds were still legally permitted to purchase

mild alcoholic beverages (de Looze et al., 2017; Schelleman-Offermans et al., 2017; Schmengler et al., 2022). Consequently, the cohort's substance use patterns may differ from both current Dutch adolescents, for whom alcohol use is less socially acceptable, and from adolescents in countries with traditionally stricter substance use norms and policies. Future research could investigate how policy environments and substance use-related societal norms moderate the relationship between risk factors (e.g., traumatic experiences) and adolescent substance use outcomes. In permissive contexts where substances are more readily available to adolescents, the link between trauma and substance use may be amplified due to increased possibilities for exposure among vulnerable youths. However, associations may also be stronger in environments where adolescent substance use is less normative, as in these contexts intensive use may serve as a stronger behavioural marker of adolescents at high risk (e.g., those with trauma histories).

6. Conclusion

This study explored whether traumatic experiences (before the age of 16) are associated with subjective and neuroendocrine stress reactivity, and whether stress reactivity measures predict substance use trajectories (tobacco, alcohol, and cannabis) from around age 16 to around age 22. We found that traumatic experiences are consistently associated with more problematic trajectories of substance use during adolescence and young adulthood and thus represent an important risk factor for the development of later substance use problems, including SUDs. However, these associations were not explained by differences in neuroendocrine or psychological stress reactivity, as associations between traumatic experiences and stress reactivity measures were absent. These findings suggest that further research on mechanisms we have not assessed may be necessary to inform interventions, such as differences in coping strategies, impulsivity, or mental health. Whilst we found no associations between subjective stress reactivity and substance use, we observed inconsistent associations between neuroendocrine stress hyporeactivity, and trajectories characterized by higher levels of substance use, with differences depending on the neuroendocrine measure and the type of substance under consideration. Our findings suggest that blunted neuroendocrine reactivity to acute stressors may serve as biomarker of increased vulnerability of problematic substance use, irrespective of traumatic experiences.

CRedit authorship contribution statement

Heiko Schmengler: Writing – review & editing, Writing – original draft, Visualization, Validation, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Hartman Catharina A:** Writing – review & editing, Project administration, Data curation. **Kristine Marceau:** Writing – review & editing, Methodology. **Matteo Giletta:** Writing – review & editing, Methodology, Funding acquisition, Conceptualization. **Margot Peeters:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition, Data curation, Conceptualization.

Data availability

Under the General Data Protection Regulation (GDPR), our dataset is considered pseudonymized rather than anonymized, and is still regarded as personal data. When participants were invited to the cohort more than 20 years ago, they were not asked to give informed consent to make their personal data publicly available in pseudonymized form. As a result of this, legal and ethical restrictions prevent the authors from making data from the TRAILS Study publicly available. Data are available upon request from the TRAILS data manager (trails@umcg.nl). Detailed information about the participation agreements with TRAILS participants is available from the ethics committee; Central Committee on Research Involving Human subjects (CCMO; tc@ccmo.nl). For more

information about accessing data from the TRAILS Study, please see <https://www.trails.nl/en/hoofdmenu/data/data-use>. The syntaxes supporting our analyses can be obtained from: <https://github.com/hschmengler/The-associations-between-traumatic-experiences-and-trajectories-of-substance-use-in-adolescence/>.

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Declaration of Competing Interest

The authors have declared that no competing interests exist.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.psyneuen.2025.107642](https://doi.org/10.1016/j.psyneuen.2025.107642).

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